

# Operating Systems Reference

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*A curated, chronologically ordered reference of notable operating systems and platforms.*

This document groups related systems into themed sections and lists them from oldest to newest within each category, alongside their official project resources.

**88 systems across 8 families** — BSD, Linux, GNU, illumos, Xen-based, experimental/research, Apple, and Windows-based custom builds. Every entry carries its first public release date, official website, source repositories, and a short overview quoting the project's own words.

**Scope and sourcing.** Descriptions quote each project's own official site or repository wherever a usable statement exists; the rest is written from those same sources. Entries are ordered by first public availability, not by rename or stable-release date — so, for example, Raspberry Pi OS is dated from Raspbian (2012) and Solus from Evolve OS (2014).

**Contributions welcome.** Corrections, missing systems, and dead links are all fair game — open an issue or a pull request. Please cite an official source for any date or quote you add.

## Table of Contents

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### I. BSD Family

- [1. NetBSD](#)
- [2. FreeBSD](#)
- [3. OpenBSD](#)
- [4. DragonFly BSD](#)
- [5. MidnightBSD](#)
- [6. GhostBSD](#)
- [7. HardenedBSD](#)

### II. Linux Distributions

- [1. Linux From Scratch](#)
- [2. Gentoo](#)
- [3. Arch Linux](#)
- [4. Puppy Linux](#)
- [5. Fedora](#)
- [6. Ubuntu](#)
- [7. Kubuntu](#)
- [8. Alpine Linux](#)
- [9. Xubuntu](#)
- [10. Linux Mint](#)
- [11. Trisquel](#)
- [12. Calculate Linux](#)

- 13. Void Linux
- 14. Tails
- 15. Mageia
- 16. Manjaro
- 17. Whonix
- 18. Kali Linux
- 19. Parrot OS
- 20. OpenMandriva
- 21. Solus
- 22. Ubuntu MATE
- 23. KDE neon
- 24. Devuan
- 25. Artix Linux
- 26. Pop!\_OS
- 27. EndeavourOS (Antergos' successor)
- 28. Ubuntu Cinnamon
- 29. GNOME OS
- 30. CachyOS
- 31. Nobara Linux
- 32. TUXEDO OS
- 33. PikaOS Linux
- 34. Vanilla OS
- 35. blendOS
- 36. Chimera Linux
- 37. Vendefoul Wolf Linux
- 38. secureblue
- 39. aerynOS
- 40. KDE Linux
- 41. Ageless Linux

#### **Additional Linux Distributions (Specialized & Niche Use Cases)**

- 42. Slackware
- 43. antiX Linux
- 44. Tiny Core Linux
- 45. Zorin OS
- 46. Parabola GNU/Linux-libre
- 47. Peppermint OS
- 48. SparkyLinux
- 49. Raspberry Pi OS
- 50. Bedrock Linux
- 51. BlackArch
- 52. MX Linux

- [53. Archman Linux](#)
- [54. RebornOS](#)
- [55. Archcraft](#)
- [56. Asahi Linux](#)

### **III. Systems Independent (GNU)**

- [1. GNU Project \(gnu.org\)](#)
- [2. GNU Guix](#)

### **IV. illumos Family**

- [1. illumos](#)
- [2. OpenIndiana](#)
- [3. SmartOS](#)
- [4. OmniOS](#)
- [5. Tribblix](#)

### **V. Xen-Based Systems**

- [1. Xen](#)
- [2. Qubes OS](#)

### **VI. Experimental & Research Systems**

- [1. FreeDOS](#)
- [2. ReactOS](#)
- [3. Haiku](#)
- [4. Redox OS](#)
- [5. SerenityOS](#)

### **VII. Apple Platforms (Operating Systems & Services)**

#### *Operating Systems*

- [1. macOS](#)
- [2. iOS Services](#)
- [3. Apple](#)
- [4. Apple Account](#)
- [5. iCloud](#)

### **VIII. Windows-Based Custom Systems (Debloated & Community Builds)**

- [1. Windows](#)
  - [2. Revision \(ReviOS\)](#)
  - [3. AtlasOS](#)
  - [4. Windows X-Lite](#)
  - [5. Tiny11](#)
  - [6. Nano11](#)
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# I. BSD Family

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*A set of Unix-like operating systems descended from the Berkeley Software Distribution, covering portable, performance-oriented, desktop, and security-hardened BSD variants.*

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## 1. NetBSD

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- **Initial Release:** Apr 1993
  - **Official Website:** <https://www.netbsd.org>
  - **Source Repository (CVS):** <https://cvsweb.netbsd.org>
  - **GitHub Mirror:** <https://github.com/NetBSD>
  - **Overview:** NetBSD — "*Of course it runs NetBSD*" — is a highly portable Unix-like operating system built around clean design and broad hardware support; for example, the same codebase runs on everything from VAX minicomputers and vintage home computers to modern ARM boards and Raspberry Pis, which is where its long-running slogan comes from in the first place.
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## 2. FreeBSD

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- **Initial Release:** Nov 1993
  - **Official Website:** <https://www.freebsd.org>
  - **Source Repository:** <https://cgит.freebsd.org>
  - **GitHub Mirror:** <https://github.com/freebsd>
  - **Overview:** FreeBSD — known for decades by its motto "*The Power to Serve*" — is a performance-oriented Unix-like system widely deployed in servers and infrastructure; for example, its networking stack and ZFS support have made it the base underneath large platforms like Netflix's content-delivery network and Sony's PlayStation operating system.
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## 3. OpenBSD

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- **Initial Release:** Jul 1996
  - **Official Website:** <https://www.openbsd.org>
  - **GitHub:** <https://github.com/openbsd>
  - **Game of Trees (Got) Mirror:** <https://openbsd.gothub.org/index.html>
  - **Overview:** OpenBSD's own homepage sums up its efforts as "*portability, standardization, correctness, proactive security and integrated cryptography*"; for example, the project performs a continuous, ongoing source-code security audit of its own codebase, and its OpenSSH tool — now the standard way most of the internet logs into remote servers — was itself born out of OpenBSD.
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## 4. DragonFly BSD

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- **Initial Release:** Jul 2003
  - **Official Website:** <https://www.dragonflybsd.org>
  - **Source Repository:** <https://gitweb.dragonflybsd.org/?p=dragonfly.git;a=summary>
  - **GitHub:** <https://github.com/DragonFlyBSD>
  - **Overview:** DragonFly BSD is a FreeBSD fork that, in the project's own words, "*provides an opportunity for the BSD base to grow in an entirely different direction*"; for example, its HAMMER2 filesystem and LWKT threading model were built from scratch by creator Matthew Dillon specifically to solve scalability problems he foresaw in FreeBSD's own approach to multiprocessing.
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## 5. MidnightBSD

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- **Initial Release:** Aug 2007
  - **Official Website:** <https://www.midnightbsd.org>
  - **GitHub:** <https://github.com/MidnightBSD>
  - **Overview:** MidnightBSD — "*a desktop operating system for x86 and x86-64 based PCs*" — is a FreeBSD derivative built specifically for everyday desktop use rather than servers; for example, it ships Xfce as its default desktop and bundles email, web browsing, and office software out of the box, filling a gap its founder felt FreeBSD itself had long overlooked.
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## 6. GhostBSD

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- **Initial Release:** Mar 2010
  - **Official Website:** <https://www.ghostbsd.org>
  - **GitHub:** <https://github.com/ghostbsd>
  - **Overview:** GhostBSD is "*a user-friendly, open-source operating system based on FreeBSD*", designed to provide a secure and efficient desktop experience for everyone; for example, it ships a ready-to-use MATE desktop out of the box and follows a slow, conservative release cycle rather than chasing bleeding-edge packages, making FreeBSD approachable for people coming from a regular Linux desktop.
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## 7. HardenedBSD

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- **Initial Release:** 2014
  - **Official Website:** <https://hardenedbsd.org>
  - **GitHub:** <https://github.com/HardenedBSD>
  - **Overview:** HardenedBSD — "*implements strong exploit mitigations and security hardening technologies on top of FreeBSD*" — began in 2013 as an Address Space Layout Randomization (ASLR) patch for FreeBSD before growing into a full security-focused fork; for example, the firewall project OPNsense adopted HardenedBSD's ASLR implementation in 2016, and by 2019 had migrated to HardenedBSD entirely.
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## II. Linux Distributions

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*A broad catalog of general-purpose, security-focused, gaming, and immutable Linux distributions, organized chronologically and linked to their official project resources.*

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### 1. Linux From Scratch

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- **Initial Release:** 1998
  - **Official Website:** <https://www.linuxfromscratch.org>
  - **Source Repository:** <https://git.linuxfromscratch.org>
  - **GitHub Mirror:** <https://github.com/lfs-book>
  - **Overview:** Linux From Scratch (LFS) — "*Your Distro, Your Rules*" — is a project that provides step-by-step instructions for building a custom Linux system entirely from source code, letting users understand and control every component of their operating system; for example, choosing their own init system, C library, or kernel configuration instead of inheriting someone else's defaults.
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### 2. Gentoo

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- **Initial Release:** 1999, 2000, 2001, Mar 2002
  - **Official Website:** <https://www.gentoo.org>
  - **Source Repository:** <https://gitweb.gentoo.org>
  - **GitHub Mirror:** <https://github.com/gentoo>
  - **Codeberg:** <https://codeberg.org/gentoo/gentoo>
  - **Overview:** Gentoo — "*can be automatically optimized and customized for just about any application or need*" — is a source-based Linux distribution built around the Portage package manager, which compiles software locally rather than installing pre-built binaries; for example, a user can toggle specific USE flags to build the very same package differently for a lightweight server versus a full gaming desktop.
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### 3. Arch Linux

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- **Initial Release:** 2001, Mar 2002
  - **Official Website:** <https://archlinux.org>
  - **GitLab:** <https://gitlab.archlinux.org/archlinux>
  - **Overview:** Arch Linux — "*a lightweight and flexible Linux distribution that tries to Keep It Simple*" — ships mostly unmodified upstream software and lets users build up their system package by package with `pacman`; for example, the community-run Arch User Repository (AUR) adds thousands of user-submitted packages on top of the official repositories.
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## 4. Puppy Linux

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- **Initial Release:** Jun 2003
  - **Official Website:** <https://puppylinux-woof-ce.github.io>
  - **GitHub:** <https://github.com/puppylinux-woof-ce>
  - **Overview:** Puppy Linux — "*a collection of multiple Linux distributions*" built on the same shared tools and principles — is designed to be ready to use, typically 1.5 GB or less in size, and customizable within minutes; for example, several official variants are built on top of Ubuntu, Debian, or Slackware packages while keeping the same lightweight Puppy-specific desktop and utilities.
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## 5. Fedora

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- **Initial Release:** Nov 2003
  - **Official Website:** <https://fedoraproject.org>
  - **Source Repository:** <https://forge.fedoraproject.org>
  - **Overview:** Fedora — sponsored by Red Hat and shaped by "*an innovative, free, and open source platform for hardware, clouds, and containers*" — ships several official editions built from the same upstream packages; for example, Fedora Workstation targets everyday desktop use while Fedora CoreOS and Fedora IoT package that same base for container hosts and embedded devices, and its immutable Fedora Atomic base — Silverblue with GNOME, Kinoite with KDE Plasma, among others — has, in turn, become a foundation other projects build on top of. Beyond the main editions, Fedora also offers official Spins that swap in a different desktop from the same repositories — KDE Plasma, Xfce, MATE, and others — and Labs, curated software bundles built for a specific field, such as the Astronomy Lab or the Security Lab, so someone in a particular niche gets a ready-to-go Fedora tailored to it from day one.
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## 6. Ubuntu

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- **Initial Release:** Oct 2004
- **Official Website:** <https://ubuntu.com>
- **Launchpad:** <https://code.launchpad.net/ubuntu>
- **GitHub:** <https://github.com/ubuntu>
- **Project Reference GitHub:** <https://github.com/canonical>
- **Overview:** Ubuntu — "*the modern, open source operating system on Linux for the enterprise*" — is a Debian-based distribution developed by Canonical, released in Desktop, Server, and Core editions; for example, the same underlying system that runs on a developer's laptop also powers unmodified instances on AWS, Azure, and Google Cloud, and embedded devices through Ubuntu Core. Its official Flavours program lets Canonical-recognized teams swap in a different desktop on that same base — Kubuntu, Xubuntu, and Ubuntu MATE among them.
- **Releases Portal:** <https://releases.ubuntu.com> Official archive of stable Ubuntu Desktop and Server ISO releases.

- **CD Image Server:** <https://cdimage.ubuntu.com> Source for official ISO images including development and alternative builds.
  - **Checksum Reference:** <https://help.ubuntu.com/community/UbuntuHashes> Documentation for verifying SHA256 checksums and signatures.
  - **Verification Guide:** <https://ubuntu.com/tutorials/how-to-verify-ubuntu> or <https://discourse.ubuntu.com/t/how-to-verify-your-ubuntu-download/14010> Official tutorial explaining how to verify downloaded Ubuntu images.
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## 7. Kubuntu

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- **Initial Release:** Apr 2005
  - **Official Website:** <https://kubuntu.org>
  - **GitHub:** <https://github.com/kubuntu-team>
  - **Overview:** Kubuntu — "*The Elegant Linux Distribution*" — is an official Ubuntu flavor that swaps the default GNOME desktop for KDE Plasma; for example, its built-in KRunner launcher can open files, convert currencies, or control music playback from one keyboard shortcut, all on top of the same package base and release schedule as standard Ubuntu.
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## 8. Alpine Linux

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- **Initial Release:** Aug 2005
  - **Official Website:** <https://alpinelinux.org>
  - **GitLab:** <https://gitlab.alpinelinux.org/alpine>
  - **GitHub:** <https://github.com/alpinelinux>
  - **Overview:** Alpine Linux — "*Small. Simple. Secure.*" — is built around musl libc and BusyBox instead of glibc and GNU coreutils, keeping a minimal container image under 8 MB; for example, its compiled binaries use position-independent execution with stack-smashing protection by default, part of why it's a common base image for Docker containers.
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## 9. Xubuntu

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- **Initial Release:** Jun 2006
  - **Official Website:** <https://xubuntu.org>
  - **GitHub:** <https://github.com/Xubuntu>
  - **Overview:** Xubuntu — "*a light, stable and configurable desktop environment with conservative workflows*" — is an Ubuntu flavor built around Xfce; for example, it keeps a traditional panel-and-menu layout and modest default resource use, making it a common pick for older or lower-specification hardware that still needs a full desktop experience.
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## 10. Linux Mint

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- **Initial Release:** Aug 2006



- **Official Website:** <https://linuxmint.com>
  - **GitHub:** <https://github.com/linuxmint>
  - **Overview:** Linux Mint — built to be "*a modern, elegant and comfortable operating system*" that's both powerful and easy to use — is developed on top of Ubuntu and Debian; for example, its own Cinnamon desktop, Update Manager, and Software Manager are all built in-house specifically to keep updates conservative and avoid the regressions that can come with more aggressive rolling releases.
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## 11. Trisquel

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- **Initial Release:** Jan 2007 (first public release); project development began in 2004
  - **Official Website:** <https://trisquel.info>
  - **GitLab:** <https://gitlab.trisquel.org/trisquel>
  - **Overview:** Trisquel — "*a 100% free operating system*" — is an Ubuntu-based, FSF-endorsed GNU/Linux distribution with every piece of proprietary software, firmware, and driver blob stripped out or replaced; for example, the project was born at the University of Vigo in Spain to serve as an educational operating system with Galician-language support, was officially presented in April 2005 with Richard Stallman's own backing, and ships MATE as its default desktop for a lighter, more traditional feel than stock Ubuntu.
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## 12. Calculate Linux

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- **Initial Release:** Jun 2007
  - **Official Website:** <https://www.calculate-linux.org>
  - **GitHub:** <https://github.com/calculatelinux>
  - **Source Repository:** <https://git.calculate-linux.org/calculate>
  - **Overview:** Calculate Linux is a Gentoo-based Linux distribution developed by Calculate Ltd., optimized for fast, ready-to-use deployment in organization environments. It layers the Calculate and Distro Portage overlays on top of Gentoo, offering Desktop, Directory Server, and Scratch editions with prebuilt binary packages while remaining fully Portage-compatible — "*a fast and functional user-friendly Linux distribution, based on Gentoo*", with a rolling-release model and a consistent, familiar desktop experience across KDE, Cinnamon, MATE, Xfce, or LXQt.
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## 13. Void Linux

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- **Initial Release:** 2008
  - **Official Website:** <https://voidlinux.org>
  - **GitHub:** <https://github.com/void-linux>
  - **Overview:** Void Linux — "*a general purpose operating system, based on the monolithic Linux kernel*" — is not a fork of any existing distribution; its `XBPS` package manager and `runit`-based init system were both written from scratch, and software is available as prebuilt binaries or built directly from source via the `xbps-src` collection.
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## 14. Tails

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- **Initial Release:** Jun 2009
  - **Official Website:** <https://tails.net>
  - **GitLab:** <https://gitlab.tails.boum.org/tails/tails>
  - **Overview:** Tails — "*a portable operating system that protects against surveillance and censorship*" — boots from a USB stick and routes all its Internet traffic through the Tor network; for example, it leaves no trace on the computer once shut down, and journalists, activists, and domestic-violence survivors have used it specifically to keep sensitive work off a shared or monitored machine.
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## 15. Mageia

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- **Initial Release:** Jun 2011
  - **Official Website:** <https://www.mageia.org>
  - **Source Repository:** <https://gitweb.mageia.org/>
  - **Overview:** Mageia — "*Stable, secure operating system for desktop & server*" — is a non-profit, community-run continuation of Mandriva Linux, formed in 2010 by former Mandriva employees and contributors; for example, its own Mageia Control Center lets a user configure hardware, networking, and system settings from one graphical tool, across a choice of KDE Plasma, GNOME, Xfce, Cinnamon, MATE, LXQt, or Enlightenment desktops.
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## 16. Manjaro

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- **Initial Release:** Jul 2011
  - **Official Website:** <https://manjaro.org>
  - **GitLab:** <https://gitlab.manjaro.org>
  - **GitHub:** <https://github.com/manjaro>
  - **Overview:** Manjaro is "*a user-friendly Linux distribution based on the independently developed Arch operating system*"; for example, it holds packages back in its own testing and stable branches before they reach users, and ships a graphical Manjaro Hardware Detection tool, both meant to smooth over the rough edges that come with using Arch directly.
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## 17. Whonix

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- **Initial Release:** Feb 2012
- **Official Website:** <https://www.whonix.org>
- **GitLab:** <https://gitlab.com/whonix>
- **GitHub:** <https://github.com/Whonix>

- **Overview:** Whonix is "*an operating system focused on anonymity, privacy and security*", built from two isolated virtual machines — a Gateway that solely runs Tor, and a Workstation that can only reach the network through it — so that even malware running as root on the Workstation cannot discover the machine's real IP address.
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## 18. Kali Linux

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- **Initial Release:** Mar 2013
  - **Official Website:** <https://www.kali.org>
  - **GitLab:** <https://gitlab.com/kalilinux>
  - **Overview:** Kali Linux, billed as "*The Most Advanced Penetration Testing Distribution, Ever*", is a Debian-based distribution built for security auditing; it ships with over 600 preinstalled tools covering tasks like vulnerability scanning, wireless attacks, and digital forensics, and is maintained by Offensive Security.
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## 19. Parrot OS

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- **Initial Release:** Apr 2013
  - **Official Website:** <https://www.parrotsec.org>
  - **GitHub:** <https://github.com/parrotsec>
  - **GitLab:** <https://gitlab.com/parrotsec>
  - **Overview:** Parrot OS is a Debian-based distribution "*designed with Security, Development, and Privacy in mind*"; for example, its Security Edition ships a full penetration-testing toolkit, while the lighter Home Edition targets everyday desktop use with privacy tools like an anonymous browsing mode built in.
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## 20. OpenMandriva

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- **Initial Release:** Nov 2013
  - **Official Website:** <https://www.openmandriva.org>
  - **Build System (ABF):** <https://abf.openmandriva.org>
  - **GitHub:** <https://github.com/OpenMandrivaAssociation>
  - **Overview:** OpenMandriva — built around "*the best balance between the most modern features and stability*", from newbies to developers — is a community-run continuation of the Mandriva Linux lineage, sponsored by the OpenMandriva Association; for example, it builds most of its packages with the Clang/LLVM toolchain rather than GCC, and offers its own `urpmi` - and `dnf` -based package management alongside a Plasma-focused desktop.
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## 21. Solus

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- **Initial Release:** May 2014 (as Evolve OS); renamed Solus in Apr 2015, with Solus 1.0 following in Dec 2015
- **Official Website:** <https://getsol.us>

- **GitHub:** <https://github.com/getsolus>
  - **Overview:** Solus is *"an operating system that is designed for home computing"*, built entirely from scratch rather than forked from Debian, Arch, or anything else; for example, its own `eo pkg` package manager follows what the project calls a *"stable-rolling"* model — delivering the newest software weekly, only after testing, rather than the bleeding-edge approach it says can sacrifice reliability for novelty — across a choice of Budgie, GNOME, Plasma, or Xfce desktops.
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## 22. Ubuntu MATE

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- **Initial Release:** Oct 2014
  - **Official Website:** <https://ubuntu-mate.org>
  - **GitHub:** <https://github.com/ubuntu-mate>
  - **Overview:** Ubuntu MATE is *"a stable, easy-to-use operating system with a configurable desktop environment"*, aimed at people who prefer a traditional desktop layout; for example, its own MATE Tweak tool can switch the entire desktop between several preset layouts — Traditional, Redmond, Cupertino, or Netbook — in a couple of clicks, without reinstalling anything.
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## 23. KDE neon

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- **Initial Release:** Jan, Jun 2016
  - **Official Website:** <https://neon.kde.org>
  - **GitLab:** <https://invent.kde.org/neon>
  - **Overview:** KDE neon — built to *"showcase KDE software exactly as the KDE developers intended it"*, with no patches — is an Ubuntu-based platform built by the KDE project itself to deliver the newest Plasma desktop and KDE application releases on top of a stable Ubuntu LTS base; for example, a bug fixed upstream in KDE software can reach neon users within days, rather than waiting for the next Kubuntu release cycle.
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## 24. Devuan

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- **Initial Release:** May 2017 (first stable release, Devuan 1.0 "Jessie"); founding announcement in Nov 2014
  - **Official Website:** <https://www.devuan.org>
  - **Source Repository:** <https://git.devuan.org/explore/repos>
  - **Overview:** Devuan was founded in November 2014 by a self-described Veteran Unix Admins collective, announced on its own site as *"a fork of Debian without systemd"* built to preserve what the project calls Init Freedom, with early administrative support from the Dyne.org non-profit foundation; that founding announcement predates any installable release by nearly three years. For example, a user can pick sysvinit, runit, or OpenRC as the init system, and can still choose Xfce, Cinnamon, KDE Plasma, LXQt, or MATE as the desktop on top.
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## 25. Artix Linux

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- **Initial Release:** Jul 2017
  - **Official Website:** <https://artixlinux.org>
  - **Source Repository:** <https://gitea.artixlinux.org/explore/repos>
  - **GitHub (former home):** <https://github.com/artix-linux>
  - **Overview:** Artix Linux — "*Simple. Fast. Systemd-free.*" — is a rolling-release, Arch-based distribution that swaps systemd for a real init system; for example, a user can pick OpenRC, runit, s6, or dinit as PID 1, on the reasoning that the first process a system starts should stay simple, secure, and stable rather than take on extra responsibilities.
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## 26. Pop!\_OS

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- **Initial Release:** Oct 2017
  - **Official Website:** <https://system76.com/pop>
  - **GitHub:** <https://github.com/pop-os>
  - **Project Reference GitHub:** <https://github.com/system76>
  - **Overview:** Pop!\_OS is billed by its maker, System76, as an operating system "*for STEM and creative professionals who use their computer as a tool to discover and create*"; for example, it ships with automatic Nvidia and AMD GPU driver handling built in and a tiling window manager that can be toggled on or off, both aimed at people doing graphics, engineering, or development work rather than general browsing.
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## 27. EndeavourOS (Antergos' successor)

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- **Initial Release:** May, Jun 2019
  - **Official Website:** <https://endeavouros.com>
  - **GitHub:** <https://github.com/endeavouros-team>
  - **Overview:** EndeavourOS picks up where Antergos left off, offering "*a lightweight Arch-based, terminal-centric system ready to personalise*"; for example, its online installer lets a user add a desktop of their choice — Xfce, KDE Plasma, GNOME, and others — on top of an otherwise close-to-vanilla Arch base, without the manual setup Arch itself requires.
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## 28. Ubuntu Cinnamon

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- **Initial Release:** Dec 2019
- **Official Website:** <https://ubuntucinnamon.org>
- **GitHub:** <https://github.com/ubuntucinnamon>

- **Overview:** Ubuntu Cinnamon — *"the most traditionally modern desktop you will ever love"* — is a community-developed, officially recognized Ubuntu flavor that pairs the Cinnamon desktop — originally built for Linux Mint — with Ubuntu's package base and release cycle; for example, someone who prefers Cinnamon's traditional panel-and-menu layout can get it with Ubuntu's hardware enablement and PPA ecosystem instead of switching distributions entirely.
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## 29. GNOME OS

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- **Initial Release:** Sep 2020 (introduced alongside GNOME 3.38.0)
  - **Official Website:** <https://os.gnome.org>
  - **GitLab:** <https://gitlab.gnome.org/GNOME/gnome-build-meta>
  - **Overview:** GNOME OS — *"producing and distributing bootable VM images for debugging and testing features"* — is a nightly, image-based reference platform built by the GNOME project itself rather than a general daily-driver distribution; for example, it rebuilds and boots a fresh system image from every commit made across GNOME's many modules, using OSTree and Flatpak, so developers, designers, and testers can try tomorrow's GNOME before it ever reaches an actual distribution's repositories.
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## 30. CachyOS

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- **Initial Release:** Jul 2021
  - **Official Website:** <https://cachyos.org>
  - **GitHub:** <https://github.com/cachyos>
  - **Overview:** CachyOS is *"an Arch Linux based distribution, which improves the performance, throughput and interactivity"* — an Arch-based distribution built around performance, using a BORE scheduler and CPU-architecture-optimized packages (x86-64-v3/v4) rather than the generic builds most distributions ship; for example, its own installer can detect a system's CPU and automatically offer the fastest matching package repository, and it ships a dedicated handheld edition for devices like the Steam Deck.
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## 31. Nobara Linux

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- **Initial Release:** Jul 2022
  - **Official Website:** <https://nobaraproject.org>
  - **GitHub:** <https://github.com/Nobara-Project>
  - **Overview:** Nobara is a Fedora-based distribution — *"A Fedora-based desktop, tuned for gaming, streaming, and content creation"* — built by Linux kernel and GNOME contributor Thomas "GloriousEggroll" Crider specifically for gaming and content creation; for example, it patches in codecs, Wine/Proton tweaks, and GPU driver fixes that stock Fedora leaves out, so games and streaming software work without the user hunting down third-party repositories first.
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## 32. TUXEDO OS

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- **Initial Release:** Sep 2022
  - **Official Website:** [https://www.tuxedocomputers.com/en/TUXEDO-OS\\_1.tuxedo](https://www.tuxedocomputers.com/en/TUXEDO-OS_1.tuxedo)
  - **GitLab:** [https://gitlab.com/tuxedocomputers/development/tuxedo\\_os](https://gitlab.com/tuxedocomputers/development/tuxedo_os)
  - **GitHub (former home):** <https://github.com/tuxedocomputers>
  - **Overview:** TUXEDO OS — "*simple by default, powerful when needed*" — is an Ubuntu-based distribution built by the hardware maker TUXEDO Computers, whose stated mission with it is making Linux accessible to the general public; it ships preinstalled on the company's own laptops and desktops, and its TUXEDO Control Center app lets a user manage fan curves, keyboard backlighting, and power profiles for their exact TUXEDO hardware from one graphical tool, something a generic Ubuntu install can't do out of the box.
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## 33. PikaOS Linux

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- **Initial Release:** Dec 2022, Jan 2023
  - **Official Website:** <https://wiki.pika-os.com/en/home>
  - **Source Repository:** <https://git.pika-os.com>
  - **GitHub (former home):** <https://github.com/PikaOS-Linux>
  - **Overview:** PikaOS is a Debian/Ubuntu-based Linux distribution — "*a gaming focused Linux distribution focussing on ease of use and high compatability*" — built by taking the gaming know-how behind Nobara — patched kernels, Wine/Proton tuning, and driver fixes aimed at running Windows games well — and pairing it with Ubuntu's broad hardware support and package base, so it works well for gaming out of the box without much manual tinkering.
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## 34. Vanilla OS

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- **Initial Release:** Oct, Nov, Dec 2022
  - **Official Website:** <https://vanillaos.org>
  - **GitHub:** <https://github.com/vanilla-os>
  - **Overview:** Vanilla OS is an immutable Ubuntu-based operating system — "*built with simplicity in mind*" — keeping the base system on an A/B root layout called ABRoot so every update is applied atomically and can be rolled back if something breaks. For example, its `apx` package manager uses OCI containers to install software from Ubuntu, Fedora, or Arch repositories side by side, without ever touching or destabilizing the immutable base.
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## 35. blendOS

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- **Initial Release:** Jan 2023
- **Official Website:** <https://blendos.co>
- **GitHub:** <https://github.com/blend-os>



- **GitLab:** <https://git.blendos.co/blendOS>
  - **Overview:** blendOS is an Arch-based immutable Linux distribution — *"a seamless blend of all Linux distributions"* — letting users install and run software from Debian, Ubuntu, Fedora, Arch, and other ecosystems side by side, with GNOME as the primary desktop alongside optional KDE Plasma or Xfce sessions. For example, a user could pull one package with `apt`, another with `dnf`, and a third with `pacman` or `yay`, all from the same shell, or boot into a full standalone desktop session from a different distro entirely — while the base system itself stays read-only and is rebuilt atomically with each update.
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## 36. Chimera Linux

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- **Initial Release:** Jun 2023 (first Alpha); project development began mid-2021
  - **Official Website:** <https://chimera-linux.org>
  - **GitHub:** <https://github.com/chimera-linux>
  - **Overview:** Chimera Linux — *"born from unhappiness with the status quo"* — is an independent, general-purpose distribution built entirely from scratch since 2021 by former Void Linux developer Nina Kolesová (q66); for example, it swaps GNU coreutils for FreeBSD's userland, uses the LLVM toolchain and musl libc instead of GCC and glibc, and runs the Dinit init system rather than systemd, while still packaging modern software like GNOME, KDE Plasma, and Flatpak on top.
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## 37. Vendefoul Wolf Linux

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- **Initial Release:** Aug, Sep 2023
  - **Official Website:** <https://vendefoul-wolf-linux.sourceforge.io>
  - **Source Repository:** <https://sourceforge.net/projects/vendefoul-wolf-linux>
  - **Overview:** Vendefoul Wolf Linux — *"a complete and free distribution available in all its versions and desktop environments"* — is a Devuan-based distribution built to work out of the box with low resource usage; for example, it ships several desktop editions and keeps telemetry out entirely, inheriting Devuan's systemd-free base rather than Debian's default one.
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## 38. secureblue

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- **Initial Release:** Dec 2023, Q1 2024
- **Official Website:** <https://secureblue.dev>
- **GitHub:** <https://github.com/secureblue/secureblue>
- **Project Reference GitHub:** <https://github.com/secureblue>
- **Overview:** secureblue — *"increase defenses against the exploitation of both known and unknown vulnerabilities"* — is a hardened, Fedora Atomic-based image built around security best practices rather than convenience; for example, it enables hardened memory allocators and stricter kernel settings by default, and ships separate images built on top of Fedora Silverblue (GNOME) and Kinoite (KDE) for users who want that hardening without assembling it themselves.



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## 39. aerynOS

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- **Initial Release:** Feb-Mar 2025
  - **Official Website:** <https://aerynos.com>
  - **GitHub:** <https://github.com/aerynOS>
  - **GitHub Forums:** <https://github.com/orgs/aerynOS/discussions>
  - **Overview:** aerynOS (formerly Serpent OS) is "*an independent performance-oriented Linux-based operating system*" that diverges significantly from traditional distributions while still aiming for a familiar, comfortable environment. Built by industry veterans and produced entirely by tooling the project developed in-house, it uses atomic updates and cutting-edge packaging in what the team describes as the next evolution in Linux distributions — an engineering-led effort built from the ground up rather than a fork of an existing base.
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## 40. KDE Linux

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- **Initial Release:** 2025 (currently Alpha)
  - **Official Website:** <https://linux.kde.org>
  - **GitLab:** <https://invent.kde.org/kde-linux/kde-linux>
  - **Overview:** KDE Linux — "*A free Linux-based operating system built by KDE*" — is an image-based distribution built from Arch Linux packages that KDE itself calls a "reference implementation" of Plasma and KDE apps, currently in Alpha; for example, it deliberately ships without the `pacman` package manager, installing software instead through Flatpak via Discover, with atomic image updates, Btrfs snapshots, and one-click rollback if something breaks.
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## 41. Ageless Linux

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- **Initial Release:** Mar 2026
  - **Official Website:** <https://agelesslinux.org>
  - **GitHub:** <https://github.com/agelesslinux/agelesslinux>
  - **Overview:** Ageless Linux is a Debian-based distribution created as an act of political protest against age verification, alongside a related hardware project building computing devices resistant to age checks; as the project puts it, "*We don't know how old you are. We don't want to know.*" It is a registered operating system under California's Digital Age Assurance Act (AB 1043, Chapter 675, Statutes of 2025, signed by Governor Newsom on October 13, 2025), and states it is in full, knowing, and intentional noncompliance with the age-verification requirements of Cal. Civ. Code § 1798.501(a). The project is run by John McCardle / FFwF Robotics LLC / Goblincorps as political commentary and civil disobedience, and is not affiliated with the Debian Project, Canonical, the Free Software Foundation, or the California State Legislature; it adds that no children were age-verified or harmed, no data was collected, and no privacy was violated in the making of the distribution — while noting that the California legislature disagrees that these things are compatible.
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## Additional Linux Distributions (Specialized & Niche Use Cases)

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### 42. Slackware

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- **Initial Release:** Jul 1993
  - **Official Website:** <https://www.slackware.com>
  - **Source Repository:** <https://git.slackbuilds.org/slackbuilds>
  - **Overview:** Slackware, the oldest still-maintained Linux distribution, is "*designed with the twin goals of ease of use and stability as top priorities*"; for example, it ships with no graphical installer and no automatic dependency resolution by default, a deliberately conservative approach that leaves package management and system configuration almost entirely in the user's hands through plain text files.
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### 43. antiX Linux

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- **Initial Release:** Jul 2007
  - **Official Website:** <https://antixlinux.com>
  - **GitHub:** <https://github.com/antiX-Linux>
  - **Overview:** antiX is "*a fast, lightweight and easy to install systemd-free Linux live CD distribution*" based on Debian Stable; for example, it can boot and run entirely from a live USB on a machine with as little as 128 MB of RAM, without ever touching the hard drive, which is part of why it's a common pick for reviving very old PCs.
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### 44. Tiny Core Linux

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- **Initial Release:** Jan 2009
  - **Official Website:** <http://www.tinycorelinux.net>
  - **Overview:** Tiny Core Linux is a minimal system built on BusyBox and FLTK that, in the project's own words, "*represents only the core needed to boot into a very minimal X desktop*" — everything beyond that arrives as user-selected extensions from an online repository; for example, it ships in three sizes: a text-only Core at roughly 17 MB, the FLWM-based TinyCore desktop at about 24 MB, and CorePlus bundling optional extras, and by default it decompresses entirely into RAM at boot rather than installing to disk, which is much of the reason it runs so quickly on hardware other distributions have long abandoned.
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### 45. Zorin OS

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- **Initial Release:** 2009
- **Official Website:** <https://zorin.com/os/>
- **GitHub:** <https://github.com/zorinos>

- **Overview:** Zorin OS is "*designed to make your computer faster, more secure and easier to use*", built on Ubuntu with a strong focus on easing the switch from Windows or macOS; for example, its built-in Zorin Appearance tool can switch the entire desktop layout to resemble Windows 11, Windows 7, or macOS with one click, easing the transition for new switchers.
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## 46. Parabola GNU/Linux-libre

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- **Initial Release:** Oct 2009
  - **Official Website:** <https://www.parabola.nu>
  - **Source Repository:** <https://git.parabola.nu>
  - **GitLab:** <https://gitlab.com/parabola>
  - **Overview:** Parabola calls itself "*a fully free, simple, and lightweight operating system*", an Arch- and Debian-based distribution built entirely from source under the GNU Free System Distribution Guidelines (FSDG); for example, its own build servers compile every package from clean chroots with networking disabled, specifically to strip out and replace any software or artwork that falls outside those guidelines, and the project earned Free Software Foundation recognition as a fully free distribution on May 20, 2011.
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## 47. Peppermint OS

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- **Initial Release:** May 2010
  - **Official Website:** <https://peppermintos.com>
  - **GitHub:** <https://github.com/peppermintos>
  - **Overview:** Peppermint OS — which, by its own account, has "*always been a minimalistic distro with signature custom tools*" — started out blending a lightweight desktop with cloud-app integration on an Ubuntu base, and is now built on Debian/Devuan Stable instead; for example, its signature Ice tool turns any website into what looks and behaves like a standalone desktop app, a feature it kept even after moving away from its original Ubuntu foundation.
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## 48. SparkyLinux

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- **Initial Release:** Oct 2011
  - **Official Website:** <https://sparkylinux.org>
  - **GitHub:** <https://github.com/sparkylinux>
  - **Overview:** SparkyLinux — "*a fast, lightweight and fully customizable operating system*" — is a Debian-based distribution offered in several editions built around different needs; for example, its GameOver edition ships preinstalled with gaming platforms and emulators, while its MinimalGUI and MinimalCLI editions strip things down to a bare Openbox desktop or command line for building a system up from scratch.
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## 49. Raspberry Pi OS

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- **Initial Release:** Jul 2012 (as Raspbian); renamed Raspberry Pi OS in May 2020
  - **Official Website:** <https://www.raspberrypi.com/software/>
  - **GitHub:** <https://github.com/raspberrypi>
  - **Overview:** Raspberry Pi OS is "*a free operating system based on Debian, optimised for the Raspberry Pi hardware*", maintained directly by Raspberry Pi itself; for example, it ships with over 35,000 precompiled packages and, unlike most desktop distributions, still supports the original 2012 Raspberry Pi Model B alongside the newest boards.
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## 50. Bedrock Linux

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- **Initial Release:** Aug 2012
  - **Official Website:** <https://bedrocklinux.org>
  - **GitHub:** <https://github.com/bedrocklinux>
  - **Overview:** Bedrock Linux is "*a meta Linux distribution which allows users to mix-and-match components*" from otherwise incompatible distributions into one broadly unified system — for example, running Debian's stable coreutils alongside Arch's cutting-edge kernel, Void's runit init system, a PDF reader kept patched through Gentoo's Portage, a font pulled from Arch's AUR, games built against Ubuntu's libraries, and business software built against CentOS's libraries, all side by side as though packaged for one single distribution. It is aimed at advanced users seeking maximum flexibility through cross-distro compatibility layers.
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## 51. BlackArch

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- **Initial Release:** Aug 2012
  - **Official Website:** <https://blackarch.org>
  - **GitHub:** <https://github.com/BlackArch>
  - **Overview:** BlackArch is "*an Arch Linux-based penetration testing distribution for penetration testers and security researchers*", with a dedicated repository holding well over 2,800 security tools at last count; for example, it can be installed as a full live system or layered on top of an existing Arch installation, adding just the BlackArch tool groups a user actually needs.
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## 52. MX Linux

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- **Initial Release:** Dec 2013
- **Official Website:** <https://mxlinux.org>
- **GitHub:** <https://github.com/MX-Linux>
- **Overview:** MX Linux is "*a cooperative venture between the antiX and former MEPIS communities*", built on a Debian Stable base with Xfce as its main desktop; for example, its own MX Tools panel bundles system snapshot, USB-installer, and driver-management utilities into one place, drawing on ideas from both antiX and the older MEPIS project.

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## 53. Archman Linux

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- **Initial Release:** Mar 2017
  - **Official Website:** <https://archman.org>
  - **GitLab:** <https://gitlab.com/Archman-OS>
  - **GitHub:** <https://github.com/archman-os>
  - **Overview:** Archman Linux — "*a fast, elegant, and user-friendly desktop experience*" — is an independent, Arch Linux-based distribution built by a community project; for example, its primary edition ships a carefully configured Xfce desktop installed through the Calamares installer, with KDE Plasma and MATE offered as alternative editions and `pamac` providing a graphical front end for package management.
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## 54. RebornOS

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- **Initial Release:** 2018
  - **Official Website:** <https://rebornos.org>
  - **GitHub:** <https://github.com/rebornos-team>
  - **Overview:** RebornOS — "*whose goal is to make Arch Linux more approachable to beginners*" — is an Arch Linux-based distribution built around a graphical installer called Reborn Creator; for example, a user can pick from a long list of desktop environments — GNOME, KDE Plasma, Xfce, Cinnamon, and others — directly during setup, arriving at a ready-to-use Arch system without touching the command line.
- 

## 55. Archcraft

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- **Initial Release:** Q4 2020
  - **Official Website:** <https://archcraft.io>
  - **GitHub:** <https://github.com/archcraft-os>
  - **Overview:** Archcraft — "*a fast, friendly, open-source operating system, for newbies, geeks & DIY enthusiasts*" — is an Arch-based distribution built around minimalist window managers rather than a full desktop environment; for example, it ships with several pre-configured Openbox themes and layouts that a user can switch between instantly from a menu, getting a polished, uncluttered look without hand-tuning dotfiles.
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## 56. Asahi Linux

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- **Initial Release:** Q4 2020, Q1 2021
- **Official Website:** <https://asahilinux.org>
- **GitHub:** <https://github.com/asahilinux>

- **Overview:** Asahi Linux *"aims to bring you a polished Linux experience on Apple Silicon Macs"*, reverse-engineering largely undocumented Apple hardware to write the needed drivers from scratch; for example, the project's own Fedora Asahi Remix now supports OpenGL and Vulkan graphics on M1 and M2 Macs, hardware Apple itself never offered official Linux drivers for.
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## III. Systems Independent (GNU)

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*A reference to the GNU operating system and its ecosystem of tools and distributions that deliver a free, Unix-like platform independent of any specific vendor or hardware line.*

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### 1. GNU Project (gnu.org)

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- **Project Announcement:** Sep 1983
  - **Development Start:** Jan 1984
  - **(FSF) Project Start:** Oct 1985
  - **GCC (GNU Compiler Collection):** 1987
  - **GNU Hurd:** 1990
  - **Official Website:** <https://www.gnu.org>
  - **GCC (GNU Compiler Collection) Project Page:** <https://gcc.gnu.org>
  - **GNU Hurd Project Page:** <https://www.gnu.org/software/hurd/>
  - **Official Source Hosting (Savannah, GNU):** <https://savannah.gnu.org>
  - **Source Hosting (Savannah, NonGNU):** <https://savannah.nongnu.org>
  - **Source Repository (GCC):** <https://gcc.gnu.org/git/gcc.git>
  - **Source Repository (Hurd):** <https://git.savannah.gnu.org/cgit/hurd>
  - **Overview:** The GNU Project has been *"developing the free Unix style operating system GNU"* since 1983, so that computer users can have the freedom to share and improve the software they use; it was announced by Richard Stallman on September 27, 1983, with development starting that following January, and its four core freedoms — to run, study, redistribute, and improve software — have since become the founding template that most free-software licenses are built around. Savannah hosts official GNU packages at <https://savannah.gnu.org>, while <https://savannah.nongnu.org> provides the same infrastructure for free software projects outside the GNU Project itself.
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### 2. GNU Guix

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- **Project Start:** Jun 2012
- **First Release:** Jan 2013
- **First Stable System Release:** May 2019
- **Official Website:** <https://guix.gnu.org>

- **Source Repository:** <https://git.guix.gnu.org>
  - **Codeberg:** <https://codeberg.org/guix/guix>
  - **Savannah (former home):** <https://cgit.git.savannah.gnu.org/cgit/guix.git>
  - **Overview:** GNU Guix — "*designed to give users more control over their general-purpose and specialized computing environments*" — is a transactional package manager and, as Guix System, a full distribution of the GNU operating system built on the Linux-libre kernel; for example, its declarative, Scheme-based package definitions let a user roll an entire system configuration back to an exact prior state with one command, the same transactional upgrade-and-rollback model it borrows from and builds upon Nix.
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## IV. illumos Family

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*A set of modern, open-source descendants of OpenSolaris that carry forward the illumos kernel into server, enterprise, and virtualization-focused operating systems.*

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### 1. illumos

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- **Initial Release:** Aug 2010
  - **Official Website:** <https://illumos.org>
  - **Source Repository:** <https://code.illumos.org>
  - **GitHub Mirror:** <https://github.com/illumos>
  - **Overview:** illumos describes itself on its own homepage as "*a Unix operating system which provides next-generation features for downstream distributions*", forked from OpenSolaris in August 2010 after Oracle discontinued that project; for example, it carries forward Solaris-era technologies like ZFS, DTrace, and Zones, and serves as the shared core — much like the Linux kernel does for Linux distributions — for every other entry in this section.
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### 2. OpenIndiana

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- **Initial Release:** Sep 2010
  - **Official Website:** <https://www.openindiana.org>
  - **GitHub:** <https://github.com/OpenIndiana>
  - **Overview:** OpenIndiana, billed on its own site as a "*Community-driven illumos Distribution*", is the most direct continuation of the old Project Indiana codename that OpenSolaris itself was born from; for example, it aims to be the de facto OpenSolaris successor installed on production servers where free security and bug fixes still matter, maintained by volunteers under the stewardship of the illumos Foundation.
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### 3. SmartOS

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- **Initial Release:** Aug 2011
  - **Official Website:** <https://tritondatacenter.com/smartos>
  - **GitHub:** <https://github.com/TritonDataCenter/smartos-live>
  - **Project Reference GitHub:** <https://github.com/TritonDataCenter>
  - **Overview:** SmartOS is billed by its maintainers as "*a lightweight container hypervisor that delivers industrial grade security*", combining OpenSolaris-derived Zones with bhyve and KVM virtualization; for example, it boots entirely into RAM rather than from disk, which makes an operating system upgrade as simple as rebooting into a newer image, and it serves as the engine behind Triton DataCenter's cloud infrastructure.
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### 4. OmniOS

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- **Initial Release:** Apr 2012
  - **Official Website:** <https://omnios.org>
  - **GitHub:** <https://github.com/omniosorg>
  - **Overview:** OmniOS describes itself on its own homepage as an "*illumos based server OS with ZFS, Bhyve, DTrace, Crossbow, SMF and Linux zone support*", deliberately built for server class hardware rather than laptops or workstations; for example, every fourth stable release is designated a Long-Term Support (LTS) build supported for three years, letting operators schedule upgrades on their own timeline instead of chasing every release.
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### 5. Tribblix

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- **Initial Release:** Oct 2012
  - **Official Website:** <https://www.tribblix.org>
  - **GitHub:** <https://github.com/tribblix>
  - **Overview:** Tribblix — "*the retro illumos distribution*" — is an independent illumos-based operating system created and still single-handedly maintained by Peter Tribble; for example, its software is installed in themed bundles called "overlays" rather than individual packages, and it deliberately keeps an old-school feel — including classic root logins and lightweight apps like AbiWord — while running modern illumos technologies such as ZFS, Zones, and DTrace underneath, on both x86 and legacy SPARC hardware.
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## V. Xen-Based Systems

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*A focused look at the Xen hypervisor and security-oriented systems built on top of it that use hardware virtualization for isolation and compartmentalization.*

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## 1. Xen

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- **Initial Release:** Oct 2003
  - **Official Website:** <https://xenproject.org/>
  - **Source Repository:** <https://xenbits.xen.org>
  - **GitHub Mirror:** <https://github.com/xen-project>
  - **Overview:** The Xen Project describes its own hypervisor as "*an open source hypervisor for infrastructure, embedded, security-sensitive, and virtualization platforms*"; for example, it is the only type-1, bare-metal hypervisor available as open source, and it powers some of the largest commercial clouds in production today alongside security-focused systems such as Qubes OS.
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## 2. Qubes OS

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- **Initial Release:** Apr 2011, Sep 2012
  - **Official Website:** <https://www.qubes-os.org>
  - **GitHub:** <https://github.com/QubesOS>
  - **Overview:** Qubes OS calls itself, with deliberate modesty, "*A reasonably secure operating system*", built around a Security by Isolation approach on top of the Xen hypervisor; for example, it lets a user run their email, browser, and work documents each in their own separate, color-coded virtual machine, so that a compromise in one — say, a malicious email attachment — never reaches the others. Edward Snowden has publicly said he uses it.
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# VI. Experimental & Research Systems

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*A selection of hobbyist, retro-compatible, and research operating systems that explore alternative kernels, architectures, and design philosophies outside mainstream platforms.*

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## 1. FreeDOS

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- **Project Start:** Jun 1994
- **First Release:** Sep 2006 (version 1.0)
- **Official Website:** <https://www.freedos.org>
- **GitHub Kernel:** <https://github.com/FDOS/kernel>
- **GitHub:** <https://github.com/FDOS>
- **GitLab:** <https://gitlab.com/FreeDOS>

- **Overview:** FreeDOS "*aims to be a complete, free, 100% MS-DOS compatible operating system*", started by Jim Hall in June 1994 the day after Microsoft announced it would stop selling MS-DOS; for example, people still use it today to run classic DOS games like DOOM on modern hardware, keep old business software running on an embedded till or cash register, or develop new DOS software from scratch.
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## 2. ReactOS

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- **Project Inception:** 1996
  - **First Release:** Feb 1998
  - **Official Website:** <https://reactos.org>
  - **GitHub:** <https://github.com/reactos>
  - **Git Mirror:** <https://git.reactos.org>
  - **Overview:** ReactOS is "*a free open source operating system based on the design principals found*" in the Windows NT architecture, built entirely from scratch with no Microsoft code involved; for example, it aims for binary compatibility with real Windows applications and drivers, so programs written for Windows Server 2003 and later can, in principle, run unmodified — though the project remains in alpha and is recommended only for testing and evaluation rather than daily use.
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## 3. Haiku

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- **Initial Release:** Aug 2001
  - **Official Website:** <https://www.haiku-os.org>
  - **Source Repository:** <https://git.haiku-os.org>
  - **GitHub Mirror:** <https://github.com/haiku>
  - **Overview:** Haiku describes itself on its own homepage as "*fast, simple to use, easy to learn and yet very powerful*", a community-driven continuation of the discontinued BeOS; for example, it reused BeOS's own Tracker file manager and Deskbar taskbar as open-sourced starting points, then rebuilt the rest of the operating system from scratch while still aiming for binary compatibility with the original BeOS API.
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## 4. Redox OS

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- **Initial Release:** Apr 2015
  - **Official Website:** <https://www.redox-os.org>
  - **GitHub:** <https://github.com/redox-os>
  - **GitLab:** <https://gitlab.redox-os.org/redox-os/redox>
  - **Overview:** Redox OS is "*a complete Unix-like microkernel-based operating system written in Rust*", created in 2015 by Jeremy Soller with safety and reliability as its central design goals; for example, its RedoxFS filesystem borrows ZFS-like features such as copy-on-write and data integrity checking, and its relibc C library lets many existing C and C++ programs be recompiled for Redox with little to no modification.
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## 5. SerenityOS

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- **First Public Release:** Oct 2018
  - **Official Website:** <https://serenityos.org>
  - **GitHub:** <https://github.com/SerenityOS>
  - **Overview:** SerenityOS calls itself "*a love letter to '90s user interfaces with a custom Unix-like core*", a graphical hobby operating system built from scratch in C++ by former Apple WebKit engineer Andreas Kling; for example, the project builds its own browser engine, window server, and userland applications entirely in-house rather than reusing existing Unix tooling, and it grew largely in public through Kling's own YouTube development videos.
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# VII. Apple Platforms (Operating Systems & Services)

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*Apple's own operating systems, alongside the core web properties and cloud services that provide identity, synchronization, and device integration across the Apple ecosystem.*

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## Operating Systems

### 1. macOS

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- **Initial Release:** Jan 1984 (as the original Macintosh System Software); modern Unix-based line since Mar 2001 (as Mac OS X)
  - **Official Website:** <https://www.apple.com/macos/>
  - **Overview:** The current release, macOS Tahoe, carries Apple's own tagline "*Fresh faced. Timelessly Mac*" — Apple's desktop and laptop operating system for the Mac, built on the Unix-based Darwin core with a hybrid XNU kernel; for example, that same Darwin foundation is open-sourced separately, while the polished desktop layer on top of it has run continuously on Mac hardware since Mac OS X's debut in 2001, through Intel and now Apple Silicon chips alike.
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### 2. iOS

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- **Initial Release:** Jun 2007 (as iPhone OS 1.0)
- **Official Website:** <https://www.apple.com/ios/>

- **Overview:** Announcing the current release, Apple's own newsroom described iOS 26 as bringing *"a beautiful new design, intelligent experiences, and improvements to the apps users rely on every day"* — Apple's mobile operating system for the iPhone, originally called iPhone OS before being renamed iOS in 2010 to reflect its spread to other Apple devices; for example, the same underlying platform was adapted into iPadOS for the iPad and shares its core with watchOS and tvOS, letting Apple ship one connected ecosystem across phones, tablets, watches, and TVs.
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## Services

### 3. Apple

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- **First Introduced:** Apr 1976
  - **Official Website:** <https://www.apple.com/>
  - **GitHub:** <https://github.com/apple>
  - **Overview:** Apple's official website — under the banner of its long-running *"Think Different"* philosophy — showcases the company's products, from iPhone and Mac to iPad, alongside its services, support resources, and official company information; for example, apple.com is also where the company still marks its own milestones, having run a "50 Years of Thinking Different" retrospective tying its 2026 anniversary back to that same 1976 campaign.
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### 4. Apple Account

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- **First Introduced:** 2000
  - **Official Website:** <https://account.apple.com/>
  - **GitHub:** <https://github.com/apple>
  - **Overview:** An Apple Account — described in Apple's own support documentation as *"the account you use to access Apple Services like the App Store"* — is the unified sign-in behind every Apple device and service; for example, signing in with the same Apple Account on an iPhone, iPad, and Mac keeps purchases, subscriptions, and settings synced across all three, and the identifier was renamed from "Apple ID" to "Apple Account" to make that central role clearer.
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### 5. iCloud

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- **Initial Release:** Oct 2011
- **Official Website:** <https://www.icloud.com>
- **GitHub:** <https://github.com/apple>
- **Overview:** iCloud's own homepage puts it simply: *"Log in to iCloud to access your photos, mail, notes, documents and more"* — Apple's cloud-based synchronization and storage platform; for example, a photo taken on an iPhone can appear moments later in the Photos app on a Mac or on iCloud.com from any browser, with edits and deletions kept in sync across every signed-in device.

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# VIII. Windows-Based Custom Systems (Debloated & Community Builds)

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*A focused Windows section highlighting official editions, popular community-tuned builds, and trusted privacy and debloat tooling for modern desktop use.*

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## 1. Windows

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- **Initial Release:** Nov 1985
  - **Official Website:** <https://www.microsoft.com/windows>
  - **Support Status:** Official Microsoft product with full commercial support.
  - **Overview:** Windows 11, per Microsoft's own announcement, "*is designed to bring you closer to what you love*" — the proprietary operating system family developed by Microsoft, with modern versions built on the Windows NT architecture; for example, the same NT-based core scales from a home laptop up to enterprise server fleets, and Windows remains the platform most PC games and business software are built against first.
  - **Optional Privacy & Debloat Tooling:**
    - [Sophia Script for Windows](#) — an open-source PowerShell module for fine-tuning and automating Windows 10 and 11 configuration, including privacy, telemetry, and UX tweaks.
    - [privacy.sexy](#) — an open-source system-hardening tool that applies transparent privacy and security scripts.
    - [Chris Titus Tech WinUtil](#) — an open-source utility to debloat, tweak, and configure Windows via a GUI.
    - [Win11Debloat](#) — a small, open-source PowerShell script to remove preinstalled apps, reduce telemetry, and declutter Windows 10 and 11.
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## 2. Revision (ReviOS)

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- **Initial Release:** 2019
- **Official Website:** <https://www.revi.cc>
- **GitHub:** <https://github.com/meetrevision>
- **Support Status:** Unofficial community Windows customization, not an official Microsoft product.

- **Overview:** ReviOS "*aspires to re-create what Windows as an operating system should have been — easy and simple*", a community-driven, debloated Windows configuration that strips unnecessary services and telemetry; for example, it disables Windows Defender and Windows Update by default (both re-enableable through its own built-in tool) to squeeze out extra performance and privacy, particularly for gaming and general desktop use.
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### 3. AtlasOS

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- **Initial Release:** 2021
  - **Official Website:** <https://atlasos.net>
  - **GitHub:** <https://github.com/Atlas-OS/Atlas>
  - **Support Status:** Community-maintained Windows optimization project, independent from Microsoft.
  - **Overview:** AtlasOS says of itself, "*Atlas brings life back to Windows, designed to maximize privacy, usability, and performance*"; it is an open-source project that applies performance, privacy, and usability optimizations to Windows using scripted "playbooks" rather than a modified installer image; for example, its AME Wizard walks a user through toggling each optimization individually, so someone can keep Windows Defender and Windows Update active while still trimming background bloat.
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### 4. Windows X-Lite

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- **Initial Release:** 2022, 2023
  - **Official Website:** <https://windowsxlite.com>
  - **Support Status:** Unofficial third-party Windows images with no Microsoft affiliation.
  - **Overview:** Windows X-Lite's own tagline describes its builds as "*designed to Enhance Performance, Privacy, Stability and Control*"; it provides unofficial lightweight Windows images with many non-essential components and preinstalled apps removed, aimed at low-spec hardware; for example, its images are built to run unmodified on everything from old, weak laptops to modern gaming desktops, old or new, weak or strong.
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### 5. Tiny11

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- **Initial Release:** Feb 2023
- **Official Website:** <https://ntdotdev.wordpress.com>
- **GitHub:** <https://github.com/ntdevlabs/tiny11builder>
- **Project Reference GitHub:** <https://github.com/ntdevlabs>
- **Support Status:** Unofficial experimental project by NTDEV, not an official Windows edition.

- **Overview:** Tiny11 is a series of heavily trimmed-down Windows 11 builds and tools created by developer NTDEV, removing many default apps and features to cut disk footprint and memory use; for example, a full Tiny11 install can fit in around 8 GB of storage against Windows 11's usual 20 GB or more, letting the same modern OS run on hardware Microsoft's official system requirements would otherwise rule out. NTDEV has since moved primary blogging to [ntdev.blog](https://ntdev.blog), though [ntdotdev.wordpress.com](https://ntdotdev.wordpress.com) remains live with the project's archive.
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## 6. Nano11

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- **Initial Release:** Sep 2025
  - **Official Website:** <https://ntdotdev.wordpress.com>
  - **GitHub:** <https://github.com/ntdevlabs/nano11>
  - **Project Reference GitHub:** <https://github.com/ntdevlabs>
  - **Support Status:** Unofficial, highly experimental project intended for advanced users only.
  - **Overview:** Nano11 is NTDEV's even more extreme follow-up to Tiny11, an experimental, minimal Windows 11 build stripped down to little more than its essential components; for example, independent testing has shrunk a Nano11 install to as little as 2.8 GB — about 3.5 times smaller than Tiny11 itself — making it intended strictly for constrained or test environments rather than everyday use.
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